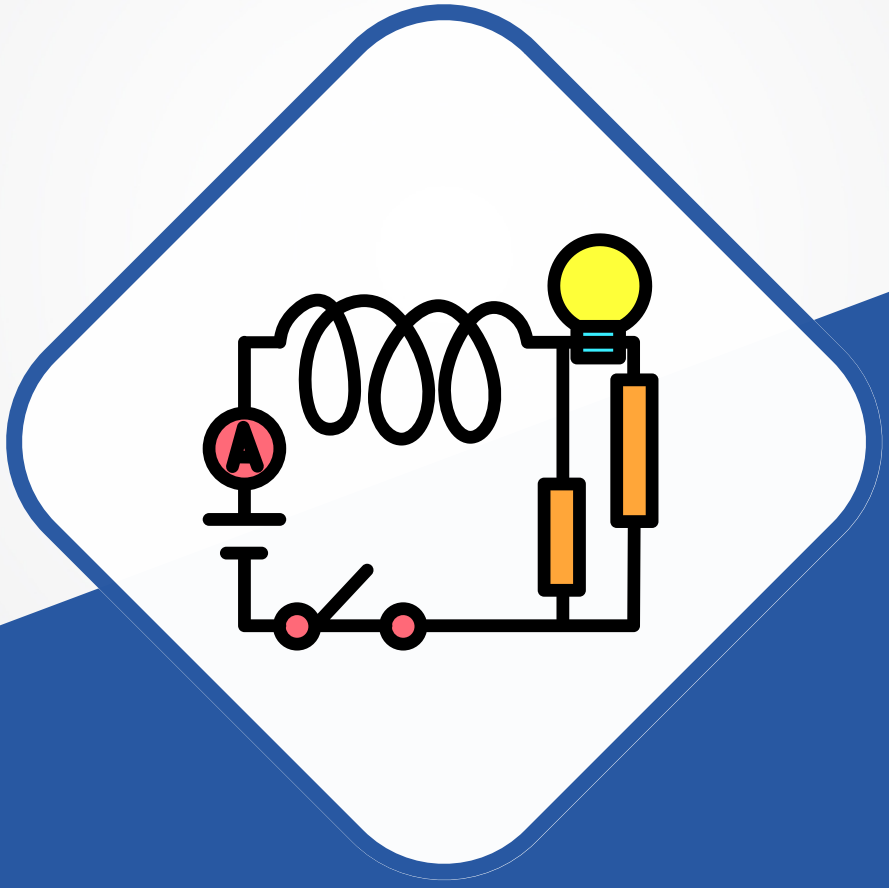
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Physics

ENTHUSIAST COURSE



STUDY MATERIAL

Current Electricity

English Medium

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CURRENT ELECTRICITY

KEY CONCEPT

1. ELECTRIC CURRENT

Any medium having practically free electric charges, free to migrate is a conductor of electricity. Metals such as gold, silver, copper, aluminium etc. are good conductors.

When charge flows in a conductor from one place to the other, then the rate of flow of charge is called electric current (I). If a charge ΔQ crosses an area in time Δt then the average electric current through the area, during this time as

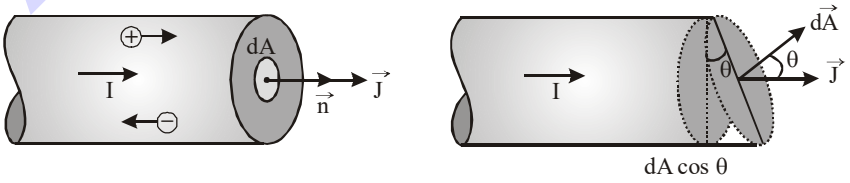
- Average current $I_{av} = \frac{\Delta Q}{\Delta t}$
- Instantaneous current $I = \underset{\Delta t \to 0}{\text{Lim}} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}$
- Current is a fundamental quantity with dimension $[M^0L^0T^0A^1]$
- Current is a scalar quantity.
- If the moving charges are positive, the current is in the direction of motion of charge. If they are negative the current is opposite to the direction of motion.



2. CURRENT DENSITY VECTOR ($\vec{j}$)

Current is a macroscopic quantity and deals with the overall rate of flow of charge through a section. To specify the current with direction in the microscopic level at a point, the term current density is introduced. Current density at any point inside a conductor is defined as a vector having magnitude equal to current per unit area surrounding that point.

- Current density at point P is given by $\vec{J} = \frac{dI}{dA}\vec{n}$



- If the cross–sectional area is not normal to the current, but makes an angle θ with the direction of current then $J = \frac{dI}{dA\cos\theta} \Rightarrow dI = JdA\cos\theta = \vec{J}.d\vec{A} \Rightarrow I = \int \vec{J} \cdot \overrightarrow{dA}$

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3. RELATION BETWEEN CURRENT DENSITY ($\vec{j}$) AND ELECTRIC FIELD ($\vec{E}$) (OHM'S LAW AT MICROSCOPIC LEVEL)

3.1 Behavior of conductor in absence of applied potential difference :

In absence of applied potential difference electrons have random motion.

The speed gained by virtue of temperature is called as thermal speed of an electron $\frac{1}{2}mv_{rms}^2 = \frac{3}{2}kt$

So thermal speed $v_{rms} = \sqrt{\frac{3kT}{m}}$ where m is mass of electron

At room temperature T = 300 K, $v_{rms} = 10^5$ m/s

- **Mean free path λ :** $(\lambda \sim 10\text{Å}) = \lambda = \frac{\text{total distance travelled}}{\text{number of collisions}}$
- **Relaxation time :** The time taken by an electron between two successive collisions is called as relaxation time τ : $(\tau \sim 10^{-14}\text{s})$,

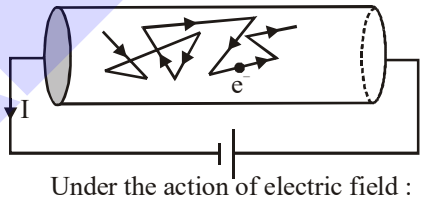
3.2 Behavior of conductor in presence of applied potential difference :

When two ends of a conductors are joined to a battery then one end is at higher potential and another at lower potential. This produces an electric field inside the conductor from point of higher to lower potential

The field exerts an electric force on free electrons causing acceleration of each electron.

Acceleration of electron $\vec{a} = \frac{\vec{F}}{m} = \frac{-e\vec{E}}{m}$

In addition to its thermal velocity, due to acceleration given by applied electric field, the electron acquires a velocity component in a direction opposite to the direction of the electric field. The gain in velocity due to the applied field is very small and is lost in the next collision.



Under the action of electric field :
Random motion of an electron
with superimposed drift

At any given time, an electron has a velocity $\vec{v}_1 = \vec{u}_1 + \vec{a}\tau_1$, where $\vec{u}_1$ = the thermal velocity and $\vec{a}\tau_1$ = the velocity acquired by the electron under the influence of the applied electric field.

τ_1 = the time that has elapsed since the last collision. Similarly, the velocities of the other electrons are $\vec{v}_2 = \vec{u}_2 + \vec{a}\tau_2, \vec{v}_3 = \vec{u}_3 + \vec{a}\tau_3, \ldots \vec{v}_N = \vec{u}_N + \vec{a}\tau_N$.

The average velocity of all the free electrons in the conductor is equal to the drift velocity $\vec{v}_d$ of the free electrons

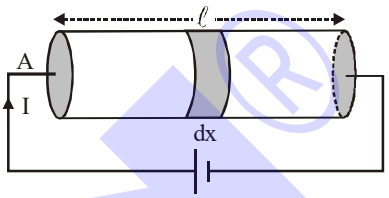
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$$\vec{v}_d = \frac{\vec{v}_1 + \vec{v}_2 + \vec{v}_3 + ...\vec{v}_N}{N} = \frac{(u_1 + \vec{a}\tau_1) + (\vec{u}_2 + \vec{a}\tau_2) + ... + (\vec{u}_N + \vec{a}\tau_N)}{N} = \frac{(\vec{u}_1 + \vec{u}_2 + ... + \vec{u}_N)}{N} + \vec{a}\left(\frac{\tau_1 + \tau_2 + ... + \tau_N}{N}\right)$$

$$\because \frac{\vec{u}_1 + \vec{u}_2 + ... + \vec{u}_N}{N} = 0 \quad \therefore \quad \vec{v}_d = \vec{a}\left(\frac{\tau_1 + \tau_2 + ... + \tau_N}{N}\right) \Rightarrow \vec{v}_d = \vec{a}\tau = -\frac{e\vec{E}}{m}\tau$$

Note : Order of drift velocity is 10^{-4} m/s.

Let the number of free electrons per unit volume in a conductor = n
Total number of electrons in dx distance = n (Adx)
Total charge dQ = n (Adx)e



Current $I = \frac{dQ}{dt} = nAe\frac{dx}{dt} = neAv_d$,

Current density $J = \frac{I}{A} = nev_d$

$$= ne\left(\frac{eE}{m}\right)\tau \because v_d = \left(\frac{eE}{m}\right)\tau \Rightarrow J = \left(\frac{ne^2\tau}{m}\right)E$$

$\Rightarrow J = \sigma E$, where conductivity $\sigma = \frac{ne^2\tau}{m}$

$$\text{resistivity} = \frac{1}{\text{conductivity}} = \rho = \frac{m}{ne^2\tau}$$

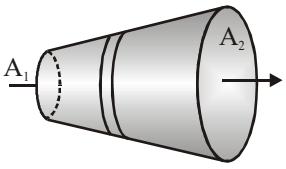
In vector form $\vec{J} = \sigma\vec{E}$ *Ohm's law (at microscopic level)*

4. RELATION BETWEEN POTENTIAL DIFFERENCE AND CURRENT (OHM'S LAW AT MICROSCOPIC LEVEL)

The current flowing through the conductor is directly proportional to the potential difference across it's two ends i.e. $I \propto V \Rightarrow V = IR$ where R is a proportionality constant, known as electric resistance.
Ohm's law (at macroscopic level)

- Ohm's law is not a universal law. The substances, which obey ohm's law are known as ohmic.

- If a steady current flows in a metallic conductor of non uniform cross section.
  Current density and drift velocity depends on area
  $I_1 = I_2$, $A_1 < A_2 \Rightarrow J_1 > J_2$, $E_1 > E_2$, $v_{d_1} > v_{d_2}$



5. RESISTANCE

The resistance of a conductor is the opposition which the conductor offers to the flow of charge.
Resistance is the property of a conductor by virtue of which it opposes the flow of current in it.
Unit : ohm, volt/ampere,
Dimension = $M L^2 T^{-3} A^{-2}$

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Resistance depends on :

- Length of the conductor ($R \propto \ell$)
- Area of cross-section of the conductor $R \propto \frac{1}{A}$
- Nature of material of the conductor $R = \frac{\rho\ell}{A}$

6. DEPENDENCE OF RESISTANCE ON TEMPERATURE

If the temperature of the conductor increases, the amplitude of the vibrations of the positive ions in the conductor also increase. Due to this, the free electrons collide more frequently with the vibrating ions and as a result, the average relaxation time decreases.

The resistance of most conductors and all pure metals increases with temperature, but there are a few in which resistance decreases with temperature. If R_o & R be the resistance of a conductor at 0º C and θº C, then it is found that $R_t = R_o (1 + \alpha \Delta\theta)$.

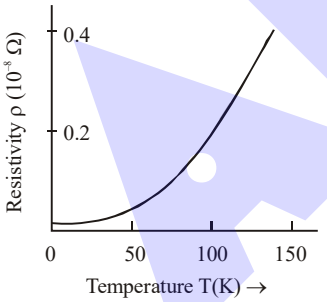
Where R_t = Resistance at t° C, R_0 = Resistance at 0° C

$\Delta\theta$ = Change in temperature, α = Temperature coefficient of resistance

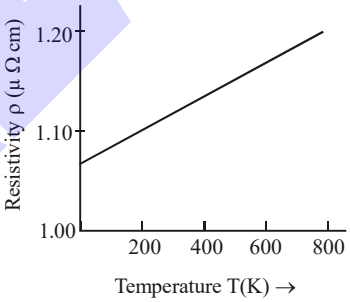
Here we assume that the dimensions of resistance do not change with temperature if expansion coefficient of material is considerable. Then instead of resistance we use same property for resistivity as $\rho = \rho_0 (1 + \alpha\Delta\theta)$

Where α is called the temperature coefficient of resistance. The unit of α is K^{-1} or $^oC^{-1}$. Reciprocal of resistivity is called conductivity and reciprocal of resistance is called conductance (G).

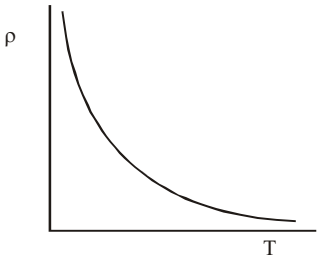
The materials for which resistance decreases with temperature, the temperature coefficient of resistance is negative.



Resistivity ρ_T of copper as a function of temperature T



Resistivity ρ_T of nichrome as a function of absolute temperature T



Temperature dependence of resistivity for a typical semiconductor.

7. KIRCHHOFF'S LAWS

I - Law (Junction law or Nodal Analysis) : This law is based on law of conservation of charge . It states that " The algebraic sum of the currents meeting at a point is zero " or total currents entering a junction equals total current leaving the junction .

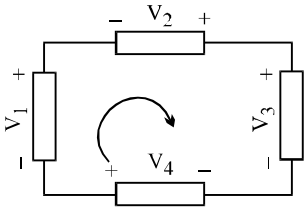
$\Sigma I_{in} = \Sigma I_{out}$. It is also known as KCL (Kirchhoff's current law).

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II - Law (Loop analysis) : The algebraical sum of all the voltages in closed circuit is zero.

$\Sigma IR + \Sigma EMF = 0$ in a closed loop . The closed loop can be traversed in any direction. While traversing a loop if higher potential point is entered, put a + ve sign in expression or if lower potential point is entered put a negative sign.

$-V_1 - V_2 + V_3 - V_4 = 0.$



Boxes may contain resistor or battery or any other element (linear or non-linear).

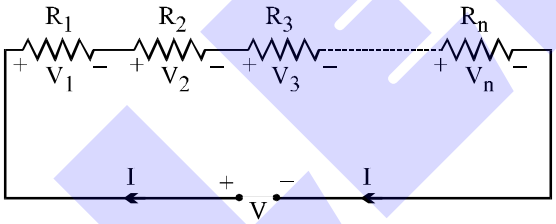
It is also known as **KVL (Kirchhoff's voltage law)** .

8. COMBINATION OF RESISTANCES

A number of resistances can be connected and all the complicated combinations can be reduced to two different types, namely series and parallel .

(i) Resistance In Series :

When the resistances are connected end to end then they are said to be in series. The current through each resistor is same. The effective resistance appearing across the battery;



$R = R_1 + R_2 + R_3 + ... + R_n$ and

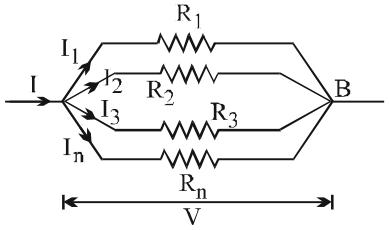
$V = V_1 + V_2 + V_3 + ... + V_n$.

The voltage across a resistor is proportional to the resistance

$$V_1 = \frac{R_1}{R_1+R_2+.....+R_n}\ V;\ V_2 = \frac{R_2}{R_1+R_2+.....+R_n}\ V;\ \text{etc.}$$

(ii) Resistance In Parallel :

A parallel circuit of resistors is one, in which the same voltage is applied across all the components in a parallel grouping of resistors $R_1, R_2, R_3, ..., R_n$.

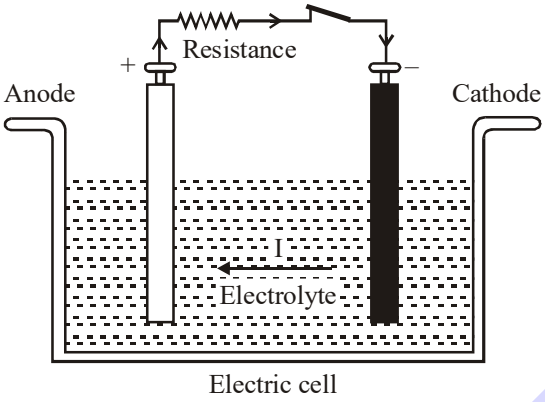


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9. EMF OF A CELL & ITS INTERNAL RESISTANCE

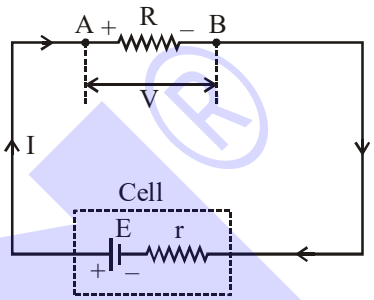
- **Electrochemical Cell :**

An electrochemical cell is a device which, by converting chemical energy into electrical energy, maintains the flow of charge in a circuit. It consists of two electrodes of different materials and an electrolyte. The electrode at higher potential is called positive terminal, or anode while at lower potential negative terminal, or cathode. When a cell is in use, i.e., discharging, current outside the cell is from anode to cathode while inside it from cathode to anode as shown in figure.



Electric cell

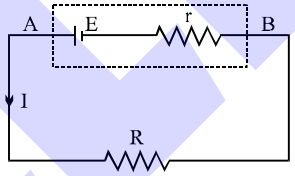
(A)



Equivalent circuit

(B)

If a cell of emf E and internal resistance r be connected with a resistance R the total resistance of the circuit is (R + r).



$I = \frac{E}{R+r}$; $V_{AB} = \frac{E}{R+r}$ where

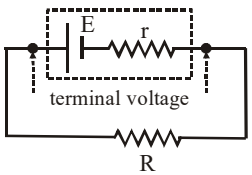
E = Terminal voltage of the battery. If $r \to 0$, cell is Ideal & $V \to E$.

Electro motive force (E. M. F.)

The potential difference across the terminals of a cell when it is not giving any current is called emf of the cell. The energy given by the cell in the flow of unit charge in the whole circuit (including the cell) is called the emf of the cell.

Terminal voltage (V)

- When current is drawn through the cell or current is supplied to cell then, the potential difference across its terminals called terminal voltage.



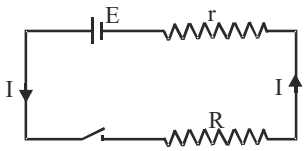
- When I current is drawn from cell, then terminal voltage is less than it's e.m.f. V = E – Ir, where r is the internal resistance of the battery

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(a) When cell is discharging :

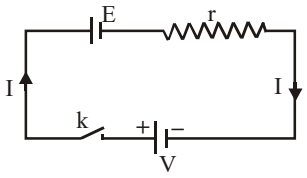
Current inside the cell is from cathode to anode.



$$\text{Current } I = \frac{E}{r+R} \Rightarrow E = IR + Ir = V + Ir \Rightarrow V = E - Ir$$

(b) When cell is charging :

Current inside the cell is from anode to cathode.



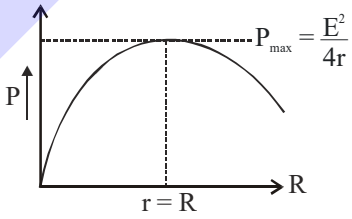
$$\text{Current } I = \frac{V-E}{r} \Rightarrow V = E + Ir$$

(c) When cell is short circuited :

$$\text{In short circuit } R = 0 \Rightarrow I = \frac{E}{R+r} = \frac{E}{r} \text{ and } V = IR = 0$$

In short circuit current from cell is maximum and terminal potential difference is zero.

(d) Power transferred to load by cell :



$$P = I^2R = \frac{E^2R}{(r+R)^2} \Rightarrow P = P_{max} \text{ if } \frac{dP}{dR} = 0 \Rightarrow r = R$$

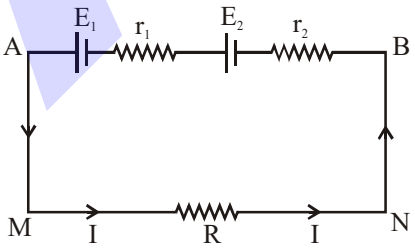
Power transferred by cell to load is maximum when

$$r = R \text{ and } P_{max} = \frac{E^2}{4r} = \frac{E^2}{4R}$$

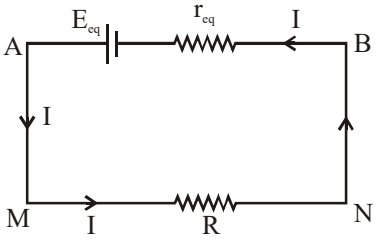
10. GROUPING OF CELLS

(i) Cells In Series :

Let two cells of emf's E_1 and E_2 and internal resistances r_1 and r_2 respectively are connected in series across a load resistance R as shown in figure.



(A)



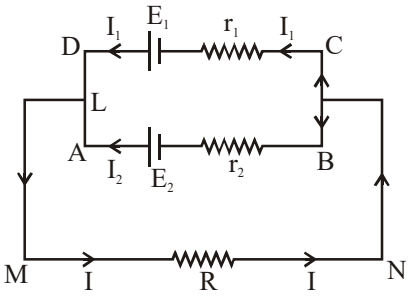
(B)

$$E_{eq} = E_1 + E_2 \text{ and } r_{eq} = r_1 + r_2$$

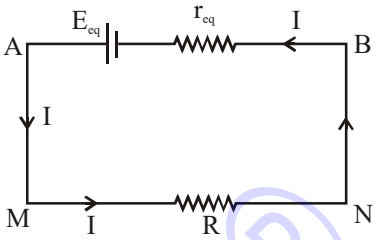
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(ii) Equivalent of emf in parallel :

Let two cells of emf's E_1 and E_2 and internal resistances r_1 and r_2 respectively are connected in parallel across a load resistance R as shown in figure.



(A)



(B)

We find that

$$r_{eq} = \left[\frac{1}{r_1} + \frac{1}{r_2}\right]^{-1} \text{ and } E_{eq} = \left[\frac{E_1}{r_1} + \frac{E_2}{r_2}\right]\left[\frac{1}{r_1} + \frac{1}{r_2}\right]^{-1}$$

i.e., $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$ and $E_{eq} = \frac{E_1 r_2 E_2 r_1}{r_1 + r_2}$

11. HEATING EFFECT OF CURRENT

- **Joules's Law of Heating**

When a current I is made to flow through a passive or ohmic resistance R for time t, heat Q is produced such that

$$Q = I^2 R t = P \times t = V I t = \frac{V^2}{R} t$$

Practical Units : 1 kilowatt hour (kWh)

$1kWh = 3.6 \times 10^6$ joule = 1 unit

- **Power :** $P = V I = \frac{V^2}{R} = I^2R$ • **SI unit :** Watt

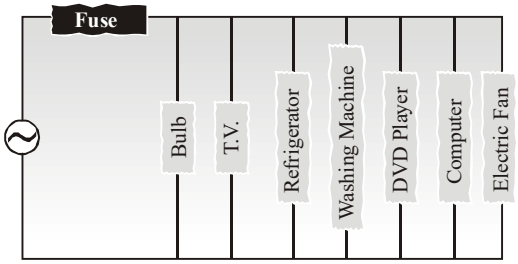
- **Rating of an appliance**

If the rating of an appliance is 50 W, 220 V, it implies that at 220 V the bulb will consume 50 W power. Hence the resistance of the appliance is $R = \frac{V^2}{P} = \frac{220^2}{50}\Omega$.

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12. FUSE WIRE

The fuse wire for an electric circuit is chosen keeping in view the value of safe current through the circuit.

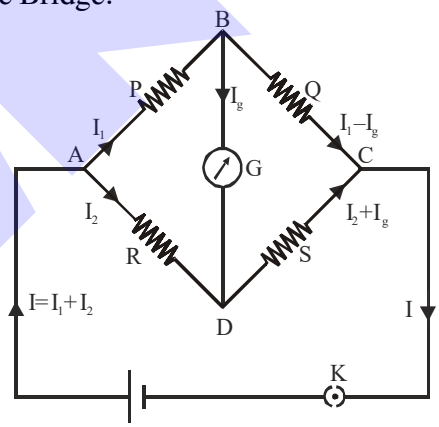


- The fuse wire should have high resistance per unit length and low melting point.
- However the melting point of the material of fuse wire should be above the temperature that will be reached on the passage of the current through the circuit

13. WHEAT STONE BRIDGE

- The configuration in the adjacent figure is called Wheat Stone Bridge.
- If current in galvanometer is zero ($I_g = 0$) then bridge is said to be balanced

$$V_D = V_B \Rightarrow I_1P = I_2R \;\&\; I_1Q = I_2S \Rightarrow \frac{P}{Q} = \frac{R}{S}$$



- If $\frac{P}{Q} < \frac{R}{S}$ then $V_B > V_D$ and current will flow from B to D.
- If $\frac{P}{Q} > \frac{R}{S}$ then $V_B < V_D$ and current will flow from D to B.

14. ELECTRICAL INSTRUMENTS

Moving coil galvanometer

(1) The instrument used to measure strength of current, by measuring the deflection of the coil due to torque produced by a magnetic field, is known as galvanometer.

(2) The total internal resistance of the galvanometer between its two terminals is called 'galvanometer resistance' and is represented by G.

(3) The current required for full scale deflection in a galvanometer is called 'full scale deflection current' and is represented by I_g (~ mA).

(4) The ratio of deflection to the current, i.e., deflection per unit current is called 'current sensitivity' of the galvanometer, i.e.,

$$CS = \frac{\theta}{I} = \frac{1}{K}$$

(5) Reciprocal of current sensitivity, i.e., current per unit deflection is called 'figure of merit', i.e.,

$$FM = \frac{I}{\theta} = \frac{1}{CS} = K$$

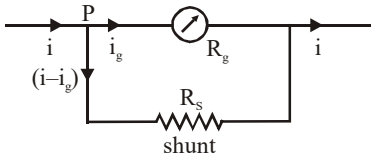
(6) Shunting a galvanometer decreases its sensitivity.

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Conversion of galvanometer into ammeter

A galvanometer can be converted into an ammeter by connecting low resistance (shunt) in parallel to its coil.

- The value of shunt resistance to be connected in parallel to galvanometer coil is given by : $R_S = \frac{R_g i_g}{i - i_g}$



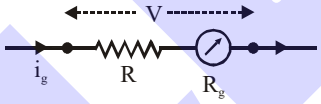
Where i = Range of ammeter

i_g = Current required for full scale deflection of galvanometer.

R_g = Resistance of galvanometer coil.

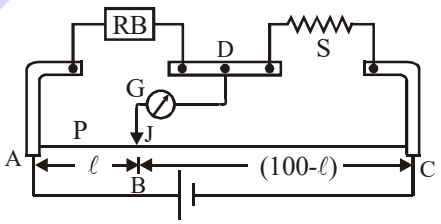
Conversion of galvanometer into voltmeter

- The galvanometer can be converted into voltmeter by connecting high resistance in series with its coil.
- The high resistance to be connected in series with galvanometer coil is given by $R = \frac{V}{i_g} - R_g$



Metre Bridge

It is based on principle of whetstone bridge. It is used to find out unknown resistance of wire. AC is 1 m long uniform wire R.B. is known resistance and S is unknown resistance. A cell is connected across 1 m long wire and Galvanometer is connected between Jockey and midpoint D. To find out unknown resistance we touch jockey from A to C and find balance condition. Let balance is at B point on wire.



AB = ℓ cm

BC = $(100 - \ell)$ cm

P = r ℓ

Q = r$(100 - \ell)$

where r = resistance per unit length on wire.

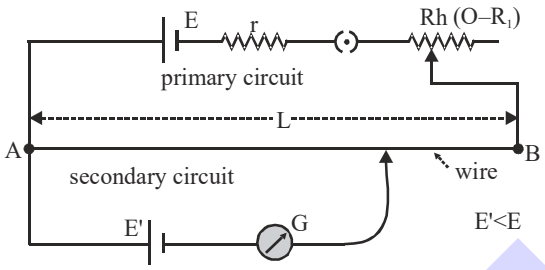
At balance condition : $\frac{P}{Q} = \frac{R}{S} \Rightarrow \frac{r\ell}{r(100-\ell)} = \frac{R}{S} \Rightarrow S = \frac{(100-\ell)}{\ell} R$

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Potentiometer

- **Working principle of potentiometer**

Any unknown potential difference is balanced on a known potential difference which is uniformly distributed over entire length of potentiometer wire. This process is named as zero deflection or null deflection method.



- **Primary circuit** contains constant source of voltage and Rheostat/Resistance Box.
- **Secondary, Unknown or galvanometer circuit**

Let ρ = Resistance per unit length of potentiometer wire

- **Potential gradient (x) (V/m)**

The fall of potential per unit length of potentiometer wire is called potential gradient.

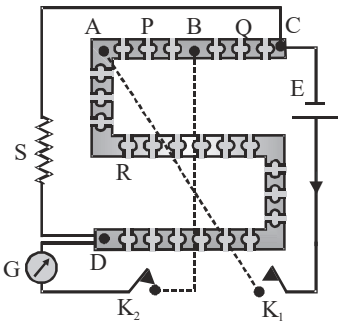
$$x=\frac{V}{L}=\frac{\text{current}\times\text{resitance of potentiometer wire}}{\text{length of potentiometer wire}}=I\left(\frac{R}{L}\right)$$

The potential gradient depends only on primary circuit and is independent of secondary circuit.

Post Office Box

It is also based on wheat stone bridge. The resistance of 10Ω, 100Ω, and 1000Ω are often connected between AB and BC. These are known as ratio arms. Resistance from 1Ω to 5000Ω are connected between A and D, this is known arm. Unknown resistance is connected between C and D.

A cell is connected between A and C with key K_1 and Galvanometer is connected between B and D with key K_2.

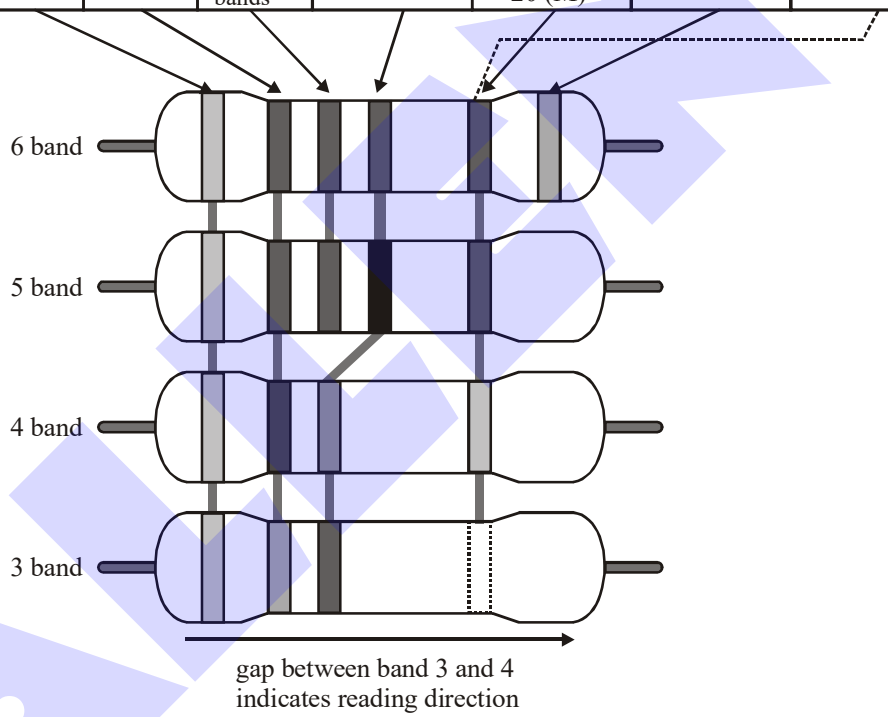


First we select ratio of resistance Q and P. For given value of S we will take value of resistance from known arm in such a way that Galvanometer show null deflection $S=\frac{Q}{P}R$. On decreasing the value of $\frac{Q}{P}$ the sensitivity of the box increases. It is used to find out the breakage in telegraph line in post and telegraph offices.

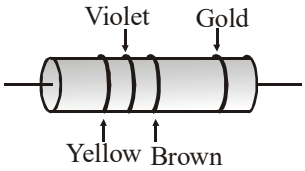
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Colour code for carbon resistors

Color	Significant figures			Multiply	Tolerance (%)	Temp. Coeff. (ppm/K)	Fail Rate (%)
black	0	0	0	× 1		250 (U)	
brown	1	1	1	× 10	1 (F)	100 (S)	1
red	2	2	2	× 100	2 (G)	50 (R)	0.1
orange	3	3	3	× 1K		15 (P)	0.01
yellow	4	4	4	× 10K		25 (Q)	0.001
green	5	5	5	× 100K	0.5 (D)	20 (Z)	
blue	6	6	6	× 1M	0.25 (C)	10 (Z)	
violet	7	7	7	× 10M	0.1 (B)	5 (M)	
grey	8	8	8	× 100M	0.05 (A)	1 (K)	
white	9	9	9	× 1G			
gold			3th digit only for 5 and 6 bands	× 0.1	5 (J)		
silver				× 0.01	10 (K)		
none					20 (M)		



Ex. What is resistance of following resistor.



Sol. Number for yellow is 4, Number of violet is 7
Brown colour gives multiplier 10^1, Gold gives a tolerance of ± 5%
So resistance of resistor is $47 \times 10^1\ \Omega \pm 5\% = 470 \pm 5\%\ \Omega$.

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